PASSENGER AIR BAG DEACTIVATION (PAD) SYSTEM [TWO-STEP DEPLOYMENT CONTROL SYSTEM]

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Purpose

- The PAD system switches the passenger's air bag and the pre-tensioner seat belt between operable (deployable) and inoperable (non-deployable).

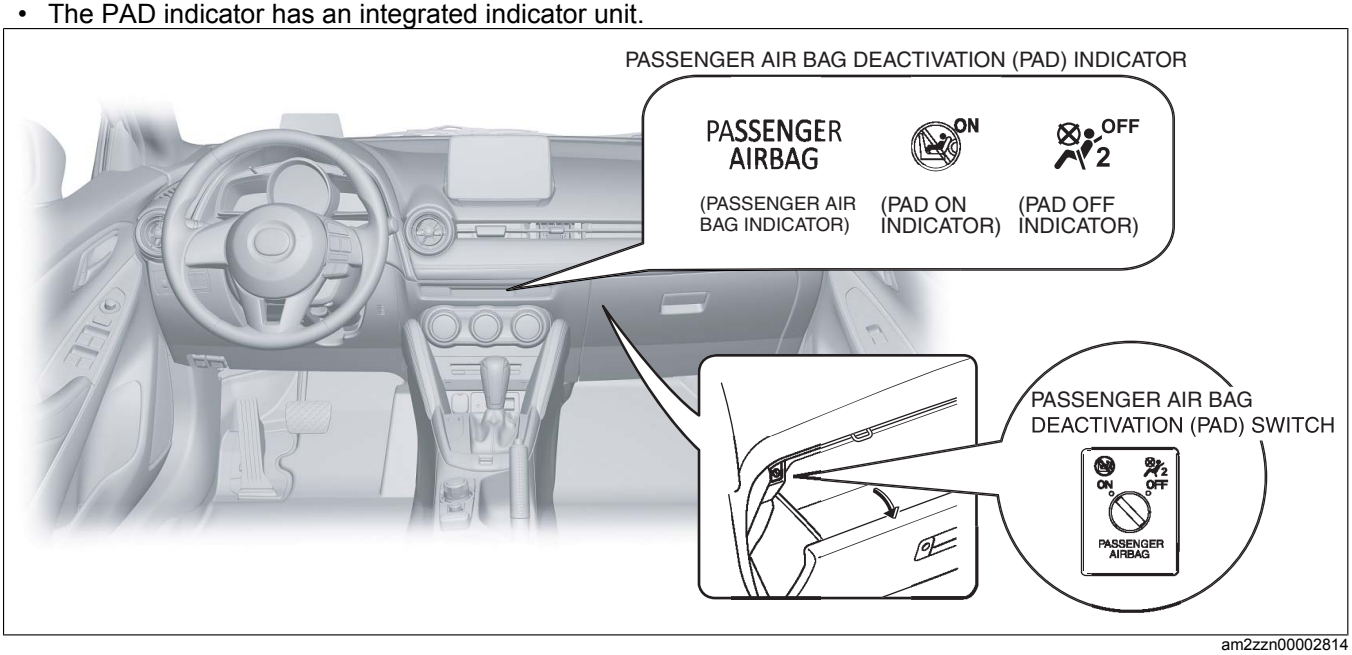
Function

- For the PAD system, the SAS control module switches the following air bag modules and the pre-tensioner seat belt between operable (deployable) and inoperable (non-deployable) based on the PAD switch status.

PAD switch status	Passenger-side air bag module/ passenger-side side air bag module/ passenger-side front pre-tensioner seat belt	PAD indicator		
		Passenger air bag indicator	PAD ON indicator	PAD OFF indicator
ON	Operable (deployable)	On	On	Off
OFF	Inoperable (non-deployable)	On	Off	On

Construction

- The PAD system consists of the PAD switch and PAD indicator.
- The PAD switch is located on the passenger-side lower panel.
- The PAD indicator has an integrated indicator unit.

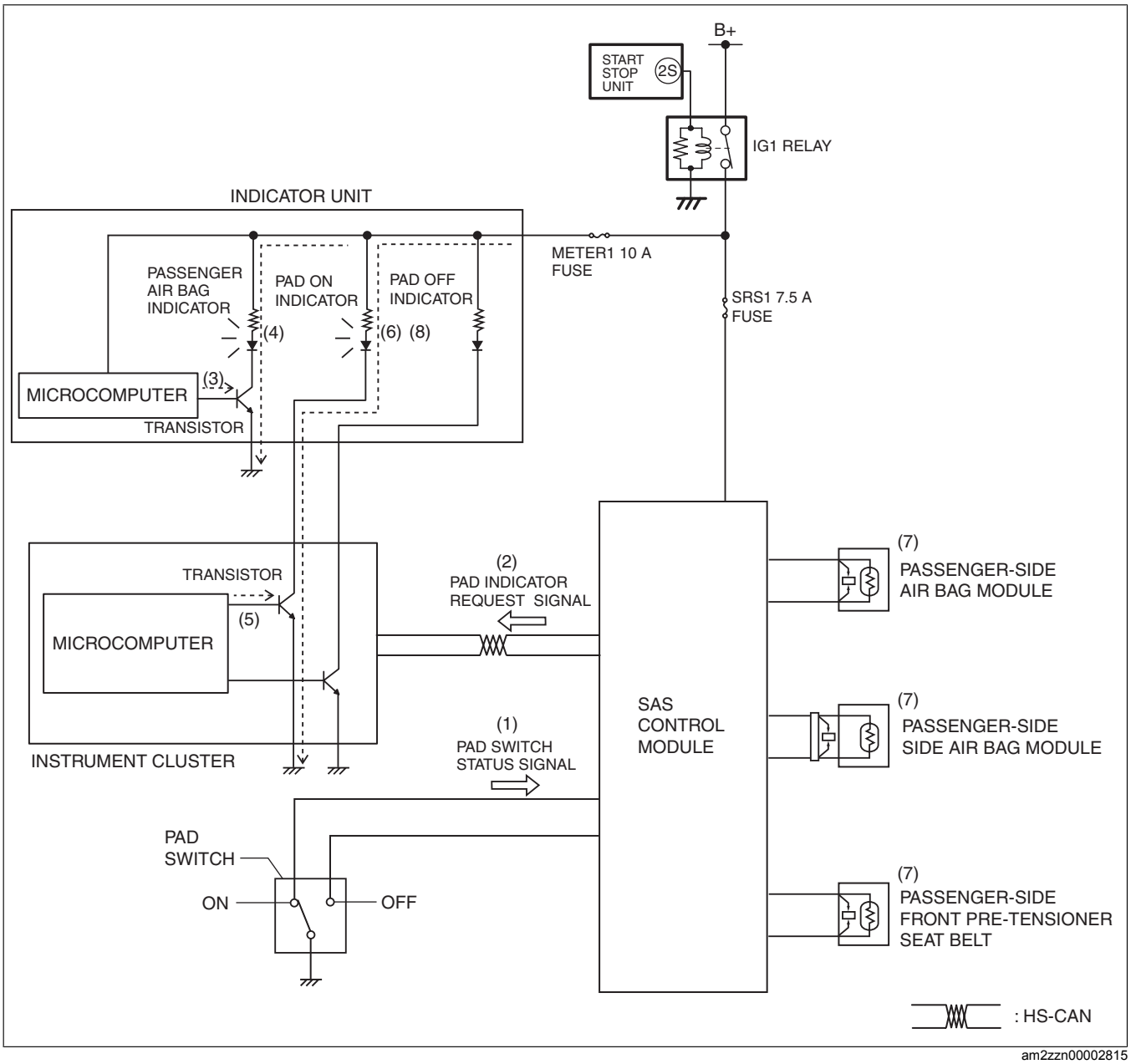


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Operation

PAD switch is ON

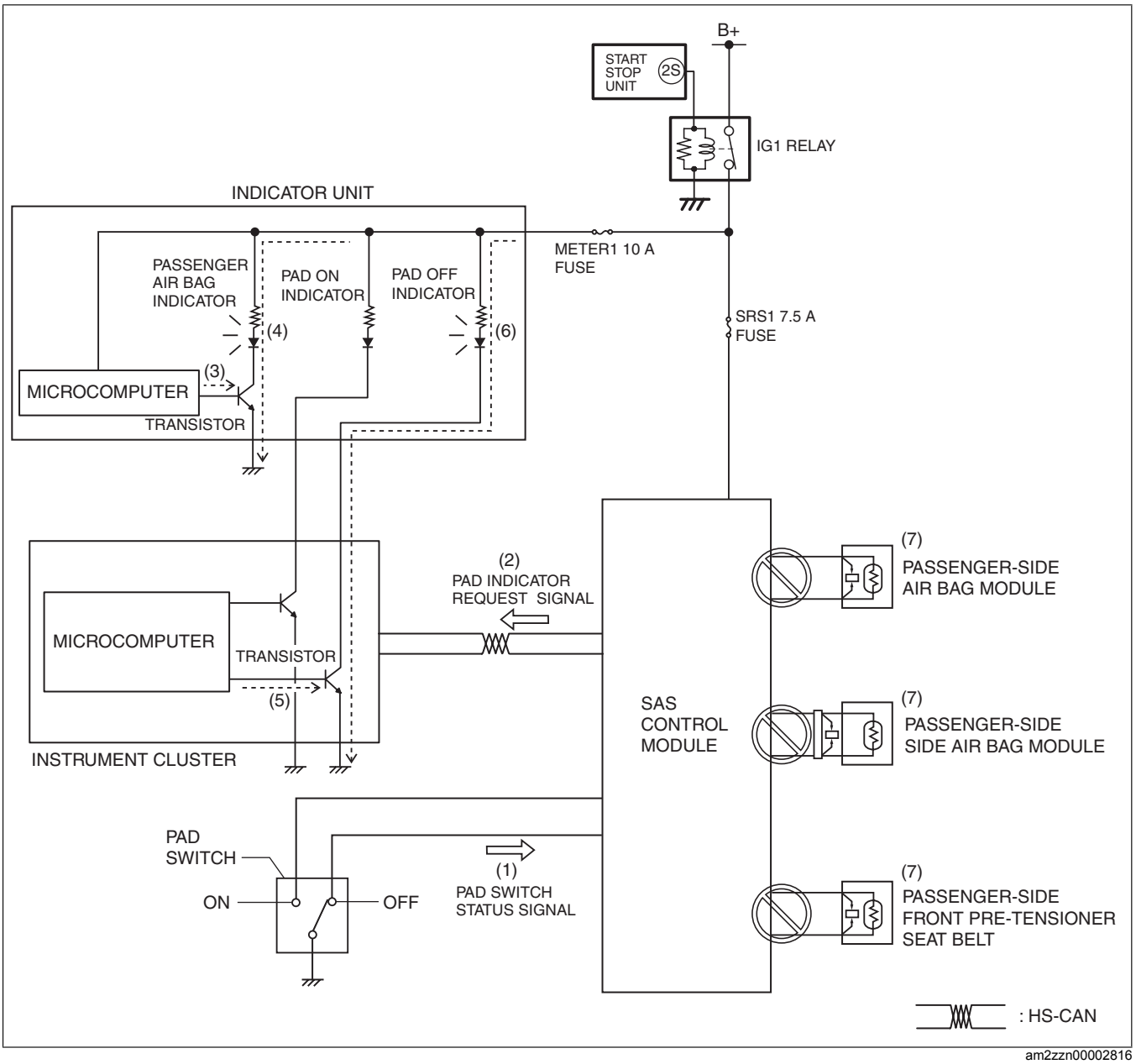
1. When the ignition is switched ON (engine off or on), the SAS control module detects (1) a PAD switch status signal and sends a PAD indicator request signal to the instrument cluster.
2. When the ignition is switched ON (engine off or on), the indicator unit turns the transistor on (3).
3. When the transistor turns on, the passenger air bag indicator turns on (4).
4. The instrument cluster turns the transistor on (5) based on the PAD indicator request signal.
5. When the transistor turns on, the PAD ON indicator turns on (6).
6. The SAS control module operates (deploys) (7) the passenger-side air bag module, passenger-side side air bag module, and passenger-side front pre-tensioner seat belt.
7. When **approx. 60 s** have elapsed after the PAD ON indicator turns on, it turns off (8).



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PAD switch is OFF

1. When the ignition is switched ON (engine off or on), the SAS control module detects (1) a PAD switch status signal and sends (2) a PAD indicator request signal to the instrument cluster.
2. When the ignition is switched ON (engine off or on), the indicator unit turns the transistor on (3).
3. When the transistor turns on, the passenger air bag indicator turns on (4).
4. The instrument cluster turns the transistor on (5) based on the PAD indicator request signal.
5. When the transistor turns on, the PAD OFF indicator turns on (6).
6. The SAS control module inhibits operation (deployment) the passenger-side air bag module, passenger-side side air bag module, and passenger-side front pre-tensioner seat belt.



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Fail-safe

- Not applicable